COVID-19 CLINICAL TRIAGE & MACHINE LEARNING PIPELINE

End-to-End Decision Architecture & MLOps Lifecycle Flowchart

Stage Input | Decision Node | Selected Path | Rejected Trap

STAGE 1: INGESTION & AUDIT

MoH Test Registry
- 278,848 swab records
- 51 days (Mar 20 - Apr 30)
- Target: 5.3% pos / 94.7% neg
- Accuracy Fallacy: 94.7% null

✔ Ground Truth Cohort
- Retain verified pos / neg
- Filter 3,892 inconclusive
- Drop 19,445 missing gender
- Clean cohort: n = 255,668
- Replicates Nature study

[SELECTED FOR BENCHMARK]

Alternative

✖ Synthetic Imputation
- Impute missing gender (7%)
- Distorts demographic data
- Real distributions smoothed
- Unnecessary: 255k is ample

[REJECTED: Model Risk]

STAGE 2: VALIDATION SPLIT

Epidemic Dynamics
- Positivity decay: 8.5% -> 6.1%
- April 15 Privacy Cliff: Age redacted to protect identities
- Temporal shift across waves

✔ Prospective Split
- Train: Mar 22-Apr 7 (98k)
- Test: Apr 8-14 (37k hold-out)
- Respects Time Arrow
- Realistic prevalence shift
- Zero look-ahead leakage

[SELECTED PROTOCOL]

Leakage Trap

✖ Random Shuffle Split
- train_test_split(shuffle=True)
- False ROC-AUC = 0.9157!
- Predicts past from future
- Catastrophic live failure

[REJECTED: Leakage Trap]

STAGE 3: PIPELINE DESIGN

8 Rapid Clinical Features
- Cough, Fever, Sore Throat
- Shortness of Breath, Headache
- Age >= 60, Gender (Male)
- Known Contact (Indication)

✔ Encapsulated Pipeline
- Scikit-Learn Pipeline
- ColumnTransformer module
- OneHotEncoder(ignore)
- Fit exclusively on Train
- Zero train-serving skew

[SELECTED ARCHITECTURE]

Engineering Risk

✖ Ad-Hoc Encoding
- Global pd.get_dummies()
- Leaks future category levels
- Breaks on novel live input
- Non-reproducible artifacts

[REJECTED: Skew Risk]

STAGE 4: MODEL BENCHMARKS

Algorithm Zoo Benchmark
- Logistic Reg: AUC 0.8336
- Decision Tree: AUC 0.8357
- Random Forest: AUC 0.8412
- 5-fold CV in Train window

✔ LightGBM Champion
- Hold-out ROC-AUC = 0.8443
- PR-AUC = 0.5448 (9x base)
- Fast leaf-wise histogram
- Matches Nature benchmark
- XGBoost: 0.8448 (parity)

[SELECTED CHAMPION]

Alternative

⚡ CatBoost & Ensembles
- CatBoost: symmetric trees
- Tiny gain (+0.0005 AUC)
- 4x computational latency
- Excessive edge complexity

[ALTERNATIVE BENCHMARK]

STAGE 5: FAIRNESS & ETHICS

Clinical Explainability
- TreeSHAP beeswarm audit
- Contact & Cough top drivers
- Subgroup disparity check: Male 0.854 vs Female 0.824

✔ Dual-Track Fallback
- Demographically Blind model
- Hold-out ROC-AUC = 0.8400
- 99.5% relative retention
- Rescues 19k unrecorded
- Immune to Apr 15 redaction

[SELECTED ETHICAL TRACK]

Fragility Trap

✖ Demographic Only
- Single model requiring age
- Total failure on Apr 15 cliff
- Age redacted by privacy laws
- Unfairly penalises young

[REJECTED: Fragility Trap]

STAGE 6: CALIBRATED TRIAGE

Clinical Scarcity
- March 2020: severe PCR limits
- Test capacity: ~3,000 / day
- Need continuous risk score
- Dynamic swab prioritisation

✔ Calibrated Cutoff
- Risk threshold tau = 0.20
- Sensitivity: 81.2% (vs 54.1%)
- Rescues 137 sick patients
- Halts community transmission
- Dynamically adjustable

[SELECTED DEPLOYMENT]

Clinical Hazard

✖ Default 0.50 Cutoff
- Default sklearn cutoff (0.50)
- Sensitivity collapses (54%)
- Misses 567 infected cases!
- Severe public health hazard

[REJECTED: Clinical Error]

OPERATIONAL 2026 DEPLOYMENT POLICY (3-TIER PRODUCTION ARCHITECTURE & GOVERNANCE):

• Tier 1 (Risk < 0.15): Safe Home Isolation | • Tier 2 (0.15 ≤ Risk ≤ 0.25): Nurse Tele-Triage & PCR Review | • Tier 3 (Risk > 0.25): Priority RT-PCR Swab
• Microservice: FastAPI Container (<5ms Inference) • Fault Tolerance: Automatic Routing to Demographically Blind Model • Governance: Daily PSI Audits